✅ 1. Competency #7: Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

- Highest relevance.
- This task is fundamentally about manipulating geometric representations under constraints:
  - Interpreting the scale (0.5 m ↔ 1 grid unit);
  - Translating real-world clearance requirements (0.5 m from walls/other sets) into discrete grid distances;
  - Working with an irregular polygonal region (the shaded seating area);
  - Modeling each table set as a point (center) with a minimum separation radius.
- Students must mentally or visually overlay a placement grid and test configurations—deep geometric manipulation in a discrete space.

✅ 2. Competency #12: Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation

- Highly relevant.
- Solving requires a clear sequence:
  1. Determine the usable interior of the shaded area (exclude a 1-grid-unit border due to wall clearance).
  2. Within this reduced region, place points (set centers) such that any two are ≥2 grid units apart.
  3. Use a systematic strategy (e.g., grid alignment, row-by-row placement) to maximize count.
  4. Verify optimality (e.g., check if shifting or rotating allows one more set).
- Missing any step (e.g., forgetting wall clearance, misjudging inter-set distance) leads to overcounting. The structured, iterative nature exemplifies Competency #12.

✅ 3. Competency #4 / #6: Make mathematical diagrams, graphs, constructions or computing artifacts; and extract mathematical information from them

- Highly relevant.
- Although the diagram is provided, students must actively use it as a dynamic workspace:
  - Many will draw circles or mark centers directly on the plan;
  - They extract spatial constraints from the visual layout (shape of seating area, proximity to walls);
  - They may construct auxiliary grids or guidelines to ensure consistent spacing.
- This transforms the static floor plan into an interactive mathematical artifact, engaging both #4 (making) and #6 (extracting).

✅ 4. Competency #13: Use an understanding of the context to guide or expedite the mathematical solving process (e.g., working to a context-appropriate level of accuracy)

- Highly relevant.
- Context informs critical modeling choices:
  - The 0.5 m clearance is not arbitrary—it reflects real human movement needs;
  - Since the grid resolution is 0.5 m, placements are naturally quantized to grid points; no need for sub-grid precision;
  - The goal is maximum integer count, not continuous optimization—so discrete, pragmatic packing suffices.
- A student who ignores context might try to place sets too close or waste space inefficiently. Context guides efficient, realistic packing.

✅ 5. Competency #8: Articulate a solution, showing and/or summarising and presenting intermediate mathematical results

- Implicitly required (though not explicitly stated as "show work," justification is expected for maximum count).
- To convince others (or oneself) that the answer is maximal, students should:
  - Describe their placement strategy;
  - Indicate where sets are located (e.g., via coordinates or markings);
  - Explain why no additional set fits (e.g., "all remaining spaces are <1 m wide").
- Even if not written, the logical verification step embodies Competency #8.

✅ 6. Competency #1: Perform a simple calculation

- Present but minimal.
- Involves basic arithmetic: counting sets, verifying distances (e.g., "this gap is only 1 grid unit → too small").
- However, the challenge is spatial, not computational—calculation supports reasoning but isn't the focus.

✅ 1. 能力 #7：操作数字、图表与统计信息、代数式、方程以及几何表示

- 最高相关性。
- 本题核心在于在约束下操作几何表示：
  - 解读比例尺（0.5 米 ↔ 1 网格单位）；
  - 将现实世界的间距要求（离墙 0.5 米、桌椅间 0.5 米）转化为离散网格中的距离规则；
  - 处理一个不规则多边形区域（阴影座位区）；
  - 将每套桌椅建模为一个点（中心），并赋予最小间隔半径。
- 学生需在脑中或图上叠加一个可放置的网格，测试不同配置——这是在离散空间中进行深度几何操作的典型体现。

✅ 2. 能力 #12：使用多步骤程序得出数学解、结论或一般化结果

- 高度相关。
- 解题需遵循清晰的步骤序列：
  1. 确定座位区的可用内部区域（排除靠近边界的 1 网格单位缓冲带）；
  2. 在该区域内，放置点（桌椅中心），确保任意两点间距 ≥ 2 网格单位；
  3. 采用系统性策略（如对齐网格、逐行放置）以最大化数量；
  4. 验证最优性（例如检查移动或旋转是否能多放一套）。
- 若遗漏任一环节（如忘记墙距、误判桌距），会导致数量虚高。这种结构化、迭代式的流程正是能力 #12 的核心。

✅ 3. 能力 #4 / #6：制作数学图表、图形、构造或计算产物；并从中提取数学信息

- 高度相关。
- 虽然图纸已提供，但学生必须将其作为动态工作平台主动使用：
  - 许多人会直接在图上画圆或标记中心点；
  - 从视觉布局中提取空间约束（座位区形状、与墙体的距离）；
  - 可能添加辅助线或参考网格以确保间距一致。
- 这使静态平面图转变为交互式数学工具，同时体现了能力 #4（制作）和 #6（提取）。

✅ 4. 能力 #13：利用对情境的理解来指导或加速数学解题过程（例如采用符合情境的精度）

- 高度相关。
- 情境理解影响关键建模决策：
  - 0.5 米的间距并非随意设定——它反映了真实的人体活动与通行需求；
  - 由于网格分辨率为 0.5 米，桌椅位置自然量化到网格交点，无需亚网格精度；
  - 目标是求最大整数套数，而非连续优化——因此采用离散、实用的布局即可。
- 忽略情境的学生可能将桌椅放得太近，或低效浪费空间。情境知识引导了高效且现实的布局策略。

✅ 5. 能力 #8：清晰表述解题过程，展示和/或总结并呈现中间数学结果

- 虽未明说“展示过程”，但隐含要求。
- 为证明答案确为最大值，学生应：
  - 描述其布局策略；
  - 标注桌椅位置（如用坐标或图示）；
  - 解释为何无法再增加一套（如“剩余空隙均小于 1 米宽”）。
- 即使未书面写出，逻辑验证环节本身就体现了能力 #8 的精神。

✅ 6. 能力 #1：执行简单计算

- 存在但次要。
- 涉及基本算术：计数、距离判断（如“此处空隙仅 1 网格单位 → 太小”）。
- 然而，挑战在于空间推理，而非计算——计算仅用于支持判断，非核心难点。